101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C 4
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

FLARE 101-1177 — 4" -FL-101-1204-A11A

3"-FL-101-1205-A11A

4"x3"

3"-WP-101-4811-B19A-IH — FROM 101-E-130A/B WASH WATER 101-1148

48"-P-101-2701-A19A-IH — FROM 101-C-101 CR COL OVHD VAP 101-1127

TW 1212 | PI 1205 | TE 1223 | TT 1223 | FF | TI 1223

NOTE-3,5 — NO POCKETS

MAKE CONNECTION AT MID POINT OF HEADER FROM TOP

MIN. — 36"-P-101-1210-A19A-IH — NOTE-3,5 — 36"x40" — 48"x40" — 40"x36" — 48"x40" — NOTE-3,5 — 36"-P-101-1211-A19A-IH — MIN.

TI 1205 | TT 1205 | TE 1205 | PI 1201 | OP | TW 1210

24"-P-101-1209-B1A-IT TO 101-E-103A/B & 105A/B — CRUDE 101-1113

NOTE-6

14"x20" — 20"x24"

FI 1201 A | FI 1201 B | FT 1201 | FF | FACING | 26"-P-101-1212-A19A-IH | SLOPE 1:500 | FE-1201 | TI 1201 | TT 1201 | TE 1201 | 12"-P-101-1205-B1A-IT

1.5E2 36.7 | PSV 1201 | 900# 2"X1-1/2" | 3/4" | LO FB | 2"-SV-101-1201-B1A/T

4 | SP 036 | NOTE-4 | B19A A19A | 3/4" | 2"-WP-101-1201-B19A-IH | NC

2"-P-101-1220-B1A-IT | MIN. | S2 | T1 | 26"x24" | 300# | 300# | T2 | S1 | A | A11A A19A IH | 3/4"

FI 1202 A | FI 1202 B | FT 1202 | FF | FACING | 26"-P-101-1215-A19A-IH | SLOPE 1:500 | FE-1202 | TI 1202 | TT 1202 | TE 1202 | 12"-P-101-1206-B1A-IT

1.5E2 36.7 | PSV 1202 | 900# 2"X1-1/2" | 3/4" | LO FB | 2"-SV-101-1202-B1A/T

SP 110 | NOTE-4 | A19A B19A | 2"-WP-101-1202-B19A-IH | 3"-FL-101-1202-A11A

2"-P-101-1221-B1A-IT | MIN. | S2 | T1 | 26"x24" | 300# | 300# | T2 | S1 | B | A11A A19A IH | 3/4" | A19A A11A IH

2" | CP 1210 | CC | SP 010 | 3/4"

FI 1203 A | FI 1203 B | FT 1203 | FF | FACING | 26"-P-101-1214-A19A-IH | SLOPE 1:500 | FE-1203 | TI 1203 | TT 1203 | TE 1203 | 12"-P-101-1207-B1A-IT

1.5E2 36.7 | PSV 1203 | 900# 2"X1-1/2" | 3/4" | LO FB | 2"-SV-101-1203-B1A/T

SP 111 | NOTE-4 | B19A A19A | 2"-WP-101-1203-B19A-IH

2"-P-101-1222-B1A-IT | MIN. | S2 | T1 | 26"x24" | 300# | 300# | T2 | S1 | C | A11A A19A IH | 3/4"

FI 1204 A | FI 1204 B | FT 1204 | FF | FACING | 26"-P-101-1215-A19A-IH | SLOPE 1:500 | FE-1204 | TI 1204 | TT 1204 | TE 1204 | 12"-P-101-1208-B1A-IT

1.5E2 36.7 | PSV 1204 | 900# 2"X1-1/2" | 3/4" | LO FB | 2"-SV-101-1204-B1A/T

SP 112 | NOTE-4 | A19A B19A | 2"-WP-101-1204-B19A-IH | 3"-FL-101-1204-A11A

2"-P-101-1223-B1A-IT | MIN. | S2 | T1 | 26"x24" | 300# | 300# | T2 | S1 | D | A19A A11A IH | 3/4"

NOTE-8 | 2" | CP 1212 | CC | SP 016 | 3/4"

14"-P-101-1200-B1A-IT

101-E-102A/B/C/D

TT 1206 | TE 1206 | TI 1206 | PI 1209 | NOTE-8 (TYP.) | MIN. | SP 008 | CC | 2" | CP 1205 | 12"-P-101-1201-B1A-IT

TE 1207 | TT 1207 | TI 1207 | PI 1206 | SP 012 | CC | 2" | CP 1206 | 12"-P-101-1202-B1A-IT

TI 1208 | TT 1208 | TE 1208 | PI 1212 | SP 014 | CC | 2" | CP 1207 | 12"-P-101-1203-B1A-IT

TE 1209 | TT 1209 | TI 1209 | PI 1215 | SP 018 | CC | 2" | CP 1208 | 12"-P-101-1204-B1A-IT

FROM 101-E-101A/B CRUDE 101-1111 — 24"-P-101-1113-B1A-IT — NOTE-6 — 16"x12" — 24"x16"

TO 101-A-101 D — TWO PHASE FLOW — NOTE-5,10 — NO POCKETS — 24"-P-101-1219-A19A-IS — CR COL OVHD 101-1128

TO 101-A-101 C — TWO PHASE FLOW — NOTE-5,10 — NO POCKETS — 24"-P-101-1218-A19A-IS — CR COL OVHD 101-1128

TO 101-A-101 B — TWO PHASE FLOW — NOTE-5,10 — NO POCKETS — 24"-P-101-1217-A19A-IS — CR COL OVHD 101-1128

TO 101-A-101 A — TWO PHASE FLOW — NOTE-5,10 — NO POCKETS — 24"-P-101-1216-A19A-IS — CR COL OVHD 101-1128

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.

THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

No.	REVISIONS	DATE	DRN	BY	CKD	APPD
4	ISSUED FOR CONSTRUCTION	10.03.22	SC	MW	YD	PV
3	ISSUED FOR CONSTRUCTION	30.09.21	SC	MW	YD	PV
2	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	02.09.20	RMB	KG	RS	SG
1	ISSUED FOR ENGINEERING/DESIGN	11.08.20	JO	KG	RS	SG
0	ISSUED FOR REVIEW	13.03.20	LC	KG	RS	SG

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
1	14.12.18	REVISED AS MARKED	PP	UA/AT	GN	AM
0	15.11.18	ISSUED FOR PACKAGE	B.K.L	UA/AT	GN	AM

PMC: इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) — ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

OWNER: एच पी सी एल राजस्थान रिफाइनरी लिमिटेड — HPCL RAJASTHAN REFINERY LTD — RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमेन्टेशन डायाग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

अनुमाप SCALE	कार्य संख्या JOB NO.	विभाग DEPT.	अनुभाग SECTN.	इकाई UNIT	आरेख संख्या DWG. No.	संशो REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. B224-CDU-TPL-101-1112 — पीसी REV. 4

DOCUMENT REVIEW DETAILS

No Comments. Doc. B224-CDU-TPL-101-1112 Rev. 4
2 Reviewed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements marked. Resubmit for review.
R Reviewed for Records. V Void.

Signature with Date: 24-AUG-2022
Name Reviewer: UMER AFZAL (C889)
Page No.: 1 of 2

NOTICE: THIS DRAWING AND THE DESIGN IT COVERS ARE THE PROPERTY OF ENGINEERS INDIA LIMITED. THEY ARE MERELY LOANED AND ON THE BORROWER'S EXPRESS AGREEMENT THAT THEY SHALL NOT BE USED, COPIED, LOANED, EXHIBITED, NOR USED EXCEPT IN THE LIMITED WAY AND PRIVATE USE PERMITTED BY ANY WRITTEN CONSENT GIVEN BY LENDER TO THE BORROWER.